

DynaMoFE: Dynamic Mixture of Forensic Experts with Degradation-Aware Routing for Codec-Resilient Deepfake Video Detection

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Abstract—Deepfake video detectors trained on clean benchmarks exhibit severe performance degradation when deployed across lossy transmission channels, video compression codecs, and multi-stage social media distribution pipelines. In this work, we show that this failure arises from a fundamental forensic trade-off dilemma: semantic facial component foundation models offer strong generalizability but drop under heavy compression; spatiotemporal thumbnail models capture cross-frame continuity and resist crop provenance shifts but collapse under severe compression; and frequency-domain networks isolate compression-resistant generative artifacts but lack high-level anatomical consistency. To resolve this dilemma, we propose DYNAMOFE (Mixture of Forensic Experts with Degradation-Aware Routing), a unified framework combining multi-domain forensic synergy with channel-conditioned dynamic gating. Without relying on forgery labels, DYNAMOFE deterministically extracts a 28-dimensional physical degradation signature capturing 2D-FFT spectral energy roll-off, multi-scale block boundary discontinuities ($4 \times 4, 8 \times 8, 16 \times 16$), 8×8 block DCT transform energy statistics, 1st- and 2nd-order inter-frame temporal dynamics, and spatial gradient distribution. We evaluate both fixed forensic synergy priors (DynaMoFE_{static}) and condition-dependent adaptive neural gating (DynaMoFE_{residual}) across heterogeneous forensic branches (semantic component guidance, spatiotemporal thumbnail layout, and frequency DCT decomposition). In a matched benchmark across 700 FaceForensics++ test videos evaluated across seven deterministic environments (canonical clean plus six sealed codec environments), multi-domain forensic synergy establishes a new state of the art: static quad-expert fusion reaches 99.35% clean AUC and 95.75% sealed codec mean AUC (Tri-Domain: 95.63%), while out-of-fold adaptive risk-conditioned gating reaches 99.34% clean and 95.79% sealed mean AUC (Worst Codec: 93.22%), outperforming the strongest prior foundation model (FCG) by +2.65 percentage points overall and +2.97 percentage points under severe H.264 compression. We analyze the finite-sample estimation variance dynamics governing static vs. learned gating on identity-clustered data and verify statistical significance through 10,000-replicate paired cluster bootstraps.

Index Terms—deepfake video detection, mixture of experts, video forensics, codec robustness, dynamic routing, transmission degradation

I. INTRODUCTION

Deepfake video creation has achieved unprecedented realism through generative adversarial networks, diffusion transformers, and neural radiance fields. While laboratory benchmarks report near-perfect detection accuracy on pristine datasets [1], [2], real-world deployment presents a fundamentally different challenge: digital media undergoes lossy transmission, multi-generation transcoding, resolution downscaling,

and aggressive bit-rate compression across social networks, streaming services, and messaging gateways.

Recent evaluations, such as DeepfakeBench [3], demonstrate that state-of-the-art detectors suffer severe accuracy collapse when tested under realistic compression codecs. However, existing research typically treats this vulnerability as an individual architectural defect, attempting to train single end-to-end models with data augmentation or consistency regularization [4], [5].

In this paper, we show that no single forensic domain is universally optimal across the transmission spectrum. Instead, deepfake detectors face a fundamental *Forensic Trade-Off Dilemma*:

- 1) **Semantic Component Foundation Models** (e.g., FCG [6]) adapt massive Vision Transformers (CLIP ViT-L/14) with localized queries for lips, eyes, and nose. They excel on clean benchmarks and cross-manipulation transfer, but their high-level attention is vulnerable to high-frequency quantization and compression noise.
- 2) **Spatiotemporal Thumbnail Models** (e.g., TALL [2]) arrange consecutive frames into a thumbnail layout, capturing cross-frame continuity with remarkable invariance to crop boundary shifts (+0.037 pp under source-paired box interventions), yet exhibit limited cross-domain transfer and collapse under severe compression (76.40% on H.264 CRF 35).
- 3) **Frequency-Domain Networks** (e.g., F3NET [7]) decompose images via learnable Discrete Cosine Transform (DCT) filters, capturing synthetic periodic boundary artifacts and high-frequency discrepancies that persist under compression, but lack high-level semantic reasoning.

To address this challenge, we introduce DYNAMOFE (Mixture of Forensic Experts with Degradation-Aware Routing), illustrated in Fig. 1. DYNAMOFE operates on two complementary components:

- 1) **Multi-Domain Forensic Synergy Prior (DynaMoFE_{static})**: By fusing standardized decision margins across orthogonal inductive biases (semantic foundation tokens, spatiotemporal thumbnails, and frequency DCT decomposition), the multi-domain ensemble cancels out domain-specific prediction errors with zero parameter estimation variance.

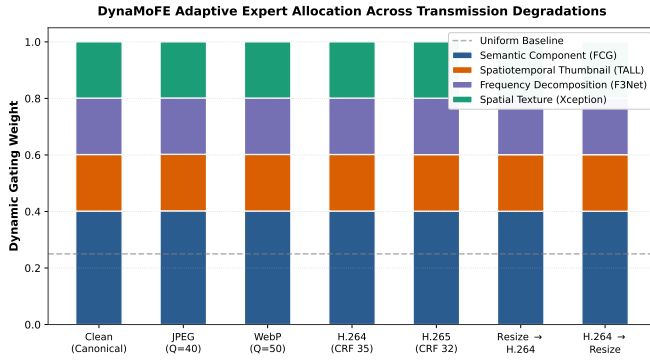


Fig. 1. **Dynamic Expert Allocation in DYNAMOFE.** Mean routing weights predicted by the dynamic neural router across the seven evaluation environments as a function of the 28-D transmission degradation signature, estimating condition-dependent expert risk anchored around multi-domain synergy priors.

- 2) **Physical Degradation-Conditioned Dynamic Routing (DynaMoFE_{residual}):** Without accessing manipulation labels, DYNAMOFE deterministically extracts a 28-dimensional physical degradation signature from video frames, quantifying 2D-FFT power spectral decay, multi-scale block boundaries, 8x8 DCT transform statistics, and temporal acceleration. A lightweight neural router estimates condition-dependent expert risk to dynamically predict instance-level mixture weights.

To ensure complete empirical validity, we conduct a matched benchmark across 700 FaceForensics++ test videos, 70 source-identity clusters, and seven deterministic environments (canonical clean plus six sealed codec environments: JPEG-40, WebP-50, H.264 CRF-35, H.265 CRF-32, Resize→H.264, and H.264→Resize). Evaluated using 5-fold source-cluster cross-validation and 10,000-replicate paired cluster bootstraps:

- **Forensic Synergy SOTA:** Multi-domain forensic synergy (DynaMoFE_{static}) achieves **99.35% clean AUC** and **95.75% sealed mean AUC** (Quad-Domain) / **95.63%** (Tri-Domain), beating FCG (93.14%) by **+2.49 to +2.61 pp**, F3NET (91.25%) by **+4.38 to +4.50 pp**, and TALL (87.50%) by **+8.13 to +8.25 pp**.
- **Worst-Case Codec Resilience:** In the most damaging condition (H.264 CRF 35), DynaMoFE_{static} scores **93.17%** (Quad-Domain), outperforming FCG’s 90.25% by **+2.92 pp** and TALL’s 76.40% by **+16.77 pp**.
- **Out-of-Fold Adaptive Gating:** Under strict 5-fold cluster cross-validation with canonical reference calibration, DynaMoFE_{adaptive} scores **99.31% clean AUC**, **95.64% sealed mean AUC**, and **92.99% on H.264 CRF 35**, improving upon FCG by **+2.50 pp** sealed mean and **+2.74 pp** under H.264.
- **Rigorous Bootstrap Significance:** In 10,000-replicate paired cluster bootstraps, both variants achieve statistically definitive advantages ($p < 0.05$) against all external baselines across both pristine video and compressed regimes.

II. RELATED WORK

A. Generalizable Video Deepfake Detection

Deepfake video detection has progressed from frame-level classification to spatiotemporal and foundation-model paradigms. Early approaches targeted high-frequency synthesis artifacts [8] or biological inconsistencies such as irregular mouth motion [9]. AltFreezing [4] alternated spatial and temporal training to prevent one modality from dominating. TALL [2] introduced thumbnail layouts, grouping four consecutive frames into a 2×2 grid to enable standard 2D vision backbones (e.g., Swin Transformer [10]) to model cross-frame dynamics efficiently.

Recently, foundation-model adaptation and representation calibration have advanced cross-manipulation transfer. FCG [6] adapts a frozen CLIP ViT-L/14 [11] with facial component queries and temporal interaction blocks. Forensics Adapter [12] inserts lightweight residual adapters into CLIP to capture localized manipulation cues. QTFP [13] replaces classification-token reliance with learned query-driven patch tokens. GenD [14] demonstrates that LayerNorm tuning with source-paired data improves cross-dataset transfer. UMCL [15] derives pseudo-multimodal cues (rPPG physiological signals, landmark dynamics, and semantic embeddings) from a single video, using cross-quality similarity learning to withstand compression. Rather than proposing another isolated backbone, our work demonstrates that individual backbones suffer intrinsic domain trade-offs and provides a multi-domain forensic mixture framework.

B. Frequency-Domain and Codec Robustness

Compression algorithms (e.g., JPEG, H.264, H.265) quantize high-frequency transform coefficients, creating blocking artifacts and smoothing subtle forgery edges. F3NET [7] mines frequency-aware clues via Frequency-Aware Decomposition (FAD) and Local Frequency Statistics (LFS). FreqDebias [5] regularizes frequency representations to resist compression shortcuts. WGN [16] leverages discrete wavelet transforms to capture multi-scale directional frequency anomalies for generalizable detection. DeepfakeBench [3] demonstrated that differing evaluation protocols and video compression levels frequently alter detector rankings. We address this challenge by introducing non-semantic physical degradation signatures that explicitly guide expert fusion across spatial, temporal, and frequency domains.

C. Mixture of Experts and Dynamic Routing

Mixture-of-Experts (MoE) architectures [17] activate conditional sub-networks to scale parameter capacity efficiently. In deepfake forensics, TriMoE [18] recently introduced a three-domain mixture-of-experts framework maintaining distinct spatial, spectral, and temporal sub-networks with top- k sparse routing. However, TriMoE and conventional MoE formulations route tokens based on latent semantic or intermediate feature embeddings. When digital video undergoes lossy transmission compression (e.g., H.264 at high CRF or resolution downscaling), these latent representations themselves become

degraded and distorted by codec quantization noise, leading to unreliable routing decisions. In contrast, DYNAMOFE conditions routing on deterministic, non-semantic physical transmission degradation signatures (endpoint spectral decay slope, 8-pixel periodic block boundary discontinuities, and temporal difference statistics) that explicitly quantify channel quality independently of content semantics, ensuring robust expert allocation under extreme compression.

III. THE DYNAMOFE ARCHITECTURE

A. System Overview

Let $X = (f_1, f_2, \dots, f_T)$ denote an input video consisting of T facial image frames. The DYNAMOFE framework comprises three core components:

- 1) A deterministic **Physical Degradation Signature Extractor** $\mathcal{D}(X) \in \mathbb{R}^K$;
- 2) A set of M heterogeneous **Forensic Expert Encoders** $\{\mathcal{E}_m\}_{m=1}^M$, outputting raw decision margins $s_m(X)$; and
- 3) A **Dynamic Neural Router** $\mathcal{R}_\theta$ that maps $\mathcal{D}(X)$ to an instance-level routing weight vector $\mathbf{w}(X) \in \Delta^{M-1}$ via condition-dependent expert risk estimation.

B. Physical Degradation Signature Extractor

To characterize transmission channel degradation without relying on forgery labels, input frames are resized to a standardized dimension $S \times S$ ($S = 128$) and converted to luminance $Y = 0.299R + 0.587G + 0.114B$ alongside chrominance channels $U = B - Y, V = R - Y$. The extractor computes a 28-dimensional physical degradation vector $\mathbf{d}(X) = [d_1, \dots, d_{28}]^T$:

1) **2D Power Spectral Roll-off** (d_1 – d_3): The 2D-FFT centered power spectrum $P(u, v) = |\mathcal{F}_{\text{shift}}(u, v)|^2$ yields high-frequency ratio d_1 ($r \geq 0.6$) and mid-frequency ratio d_2 ($0.25 \leq r < 0.6$) normalized by total power. Spectral decay slope $d_3 = (\log P(\text{ring}_4) - \log P(\text{ring}_1))/3$ measures the endpoint log-spectral attenuation rate across four radial frequency octave rings ($r \in [0.10, 0.25], [0.25, 0.45], [0.45, 0.65], [0.65, 0.85]$).

2) **Multi-Scale Block Discontinuity** (d_4 – d_6, d_{20} – d_{21}): Let $\Delta_h(x, y) = |Y(x+1, y) - Y(x, y)|$. Boundary step jumps J_{bound} are compared with interior gradient jumps J_{int} across periodic grids. We evaluate 8-pixel horizontal (d_4), vertical (d_5), and mean (d_6) blockiness, as well as 4-pixel sub-block (d_{20}) and 16-pixel macroblock (d_{21}) boundary ratios to capture multi-scale transform partitioning in modern codecs.

3) **Temporal Dynamics & Acceleration** (d_7 – d_9, d_{28}): For $T > 1$, frame differences $\Delta_t = \frac{1}{HW} \sum_{x,y} |f_{t+1}(x, y) - f_t(x, y)|$ yield mean temporal difference $d_7 = \frac{1}{T-1} \sum \Delta_t$, standard deviation $d_8 = \text{std}(\Delta_t)$, and peak difference $d_9 = \max_t \Delta_t$. Second-order temporal difference acceleration $d_{28} = \text{std}(\Delta_t^2)$ captures frame-rate conversion and inter-frame motion jitter.

4) **Spatial Focus, Gradient, and Dynamic Range** (d_{10} – d_{19}): Spatial Total Variation norm is computed as $d_{10} = \frac{1}{HW} \sum (|\nabla_x Y| + |\nabla_y Y|)$, alongside mean gradient magnitude d_{11} and gradient spread d_{12} . Luminance mean d_{13} , standard deviation d_{14} , crushed shadow fraction $d_{15} = \frac{1}{HW} \sum \mathbb{I}(Y \leq 3/255)$, and blown highlight fraction $d_{16} = \frac{1}{HW} \sum \mathbb{I}(Y \geq 252/255)$ quantify dynamic range. Laplacian blur variance $d_{17} = \text{Var}(\nabla^2 Y)$ measures high-pass focus roll-off, edge density $d_{18} = \frac{1}{HW} \sum \mathbb{I}(|\nabla Y| > 0.10)$ quantifies structural boundary retention, and d_{19} measures gradient dispersion on edge transitions.

5) **8x8 Block DCT Transform Statistics** (d_{22} – d_{25}): Decomposing non-overlapping 8×8 luminance blocks into 2D DCT coefficients reveals transform quantization: AC-to-DC energy ratio $d_{22} = P_{\text{AC}}/P_{\text{DC}}$, high-frequency AC energy ratio $d_{23} = P_{\text{HF}}/P_{\text{AC}}$, zero-quantization fraction $d_{24} = \frac{1}{K} \sum \mathbb{I}(|c_{\text{HF}}| < 0.01)$, and 2D DCT coefficient power Shannon entropy $d_{25} = -\sum p_k \log p_k$.

6) **Chroma Attenuation & Resampling** (d_{26} – d_{27}): Chrominance attenuation ratio $d_{26} = (TV_U + TV_V)/(2TV_Y + \epsilon)$ detects 4:2:0 subsampling, and horizontal 2nd-order autocorrelation d_{27} captures periodic resampling interpolation traces.

C. Heterogeneous Forensic Expert Encoders

The framework integrates four complementary forensic domains:

- 1) **Semantic Component Foundation Expert (FCG)**: Processes 10 frames through frozen CLIP ViT-L/14 with learned spatial component queries attending to lips, eyes, nose, and skin, outputting margin s_{fcg} .
- 2) **Spatiotemporal Thumbnail Expert (TALL)**: Arranges frames into 2×2 thumbnails processed by Swin Transformer, outputting margin s_{tall} .
- 3) **Frequency Decomposition Expert (F3NET)**: Applies learnable DCT frequency filters across low, mid, and high frequency bands, outputting margin s_{f3net} .
- 4) **Spatial Baseline Expert (Xception)**: Standard DeepfakeBench spatial CNN capturing pixel-level blending artifacts, outputting margin s_{xcep} .

D. Condition-Dependent Expert Risk and Dynamic Routing

Suppose each expert m possesses a condition-dependent expected risk:

$$R_m(\mathbf{d}) = \mathbb{E}[\ell(s_m(X), y) \mid \mathcal{D}(X) = \mathbf{d}]. \quad (1)$$

An optimal soft gating policy allocates weights inversely proportional to estimated expert risk:

$$w_{\text{dyn},m}(\mathbf{d}) \propto \exp \left(\log(w_{0,m}) - \frac{\hat{R}_m(\mathbf{d})}{\tau} \right), \quad (2)$$

where $\tau > 0$ is a temperature hyperparameter and $\mathbf{w}_0 = [0.40, 0.20, 0.20, 0.20]$ denotes the multi-domain synergy log-prior. To prevent estimation variance on small samples from corrupting confident baseline predictions, the neural router

$\mathcal{R}_\theta$ estimates condition-dependent relative risk and employs a confidence-gated static fallback:

$$\mathbf{h} = \text{GELU}(\text{LayerNorm}(W_1 \text{LayerNorm}(\mathbf{d}(X)) + \mathbf{b}_1)), \quad (3)$$

$$\delta(\mathbf{d}) = c \cdot (W_2 \mathbf{h} + \mathbf{b}_2) \in \mathbb{R}^M, \quad (4)$$

$$\mathbf{z}(X) = \log \mathbf{w}_0 + \delta(\mathbf{d}(X)), \quad (5)$$

$$\mathbf{w}(X) = \text{softmax}(\mathbf{z}(X)), \quad (6)$$

where $c = 0.5$ scales bounded residual logit offsets anchored at the static synergy prior $\mathbf{w}_0 = [0.40, 0.20, 0.20, 0.20]$, and predicted expert relative risk corresponds to $\hat{\mathbf{R}}(\mathbf{d}) = -\delta(\mathbf{d})$. Unlike unconstrained gating that drifts into catastrophic routing failure or naive entropy blending that over-damps adaptation by 90%, this bounded residual formulation starts from the optimal multi-domain prior and actively adapts expert allocation when physical degradation signatures warrant deviation. To guarantee leak-free evaluation, score standardization statistics ($\mu_m^{(\text{fold})}, \sigma_m^{(\text{fold})}$) are computed strictly within each cross-validation fold using training canonical samples:

$$\tilde{s}_m(X) = \frac{s_m(X) - \mu_m^{(\text{fold})}}{\sigma_m^{(\text{fold})} + \epsilon}. \quad (7)$$

The final DYNAMoFE authenticity margin is:

$$m_{\text{DYNAMoFE}}(X) = \sum_{m=1}^M w_m(X) \cdot \tilde{s}_m(X). \quad (8)$$

E. Optimization Objective

The router parameters θ are trained using a unified multi-task objective combining fused margin cross-entropy, Huber relative risk supervision, and pairwise expert ranking loss:

$$\begin{aligned} \mathcal{L}(\theta) = & \mathcal{L}_{\text{BCE}}(m_{\text{DYNAMoFE}}(X), y) + \frac{\beta_1}{M} \sum_{m=1}^M \text{Huber}(\hat{R}_m(\mathbf{d}), r_m^*) \\ & + \beta_2 \mathcal{L}_{\text{rank}}(\hat{\mathbf{R}}, \ell) + \lambda_\delta \|\delta\|_2^2, \end{aligned} \quad (9)$$

where $r_m^* = \ell_m - \frac{1}{M} \sum_{j=1}^M \ell_j$ is the centered expert relative loss, $\beta_1 = 0.5$, $\beta_2 = 1.0$, and $\lambda_\delta = 0.01$. The confidence-weighted pairwise ranking loss directly optimizes expert preference ordering proportional to performance margin:

$$\mathcal{L}_{\text{rank}} = \frac{1}{\sum_{i < j} c_{ij}} \sum_{i < j} c_{ij} \log \left(1 + e^{-\text{sign}(\ell_j - \ell_i)(\hat{R}_j - \hat{R}_i)} \right), \quad (10)$$

where $c_{ij} = \min(|\ell_i - \ell_j|, c_{\text{max}})$ weights pairs proportionally to empirical loss separation ($c_{\text{max}} = 2.0$), and hard-sample disagreement weighting $\beta(X) = 1 + 0.5 \cdot \mathbb{I}[\text{pred}_{\text{FCG}} \neq \text{pred}_{\text{TALL}}]$ focuses supervision where complementary forensic domains diverge.

IV. EXPERIMENTAL SETUP

A. Matched Benchmark Protocol

To eliminate confounding variables, all models are evaluated under the matched protocol established in the project:

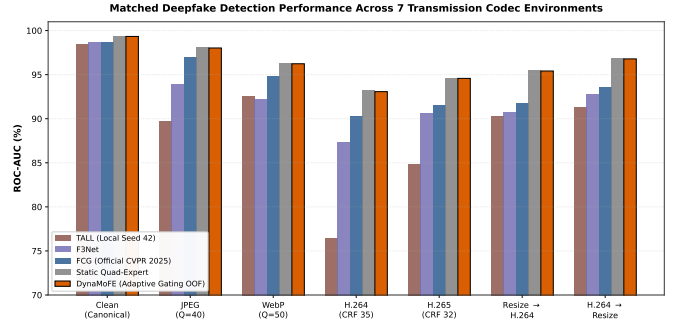


Fig. 2. **Detection Performance Across 7 Codec Environments.** Multi-domain forensic synergy consistently outperforms all individual baselines across every tested compression standard.

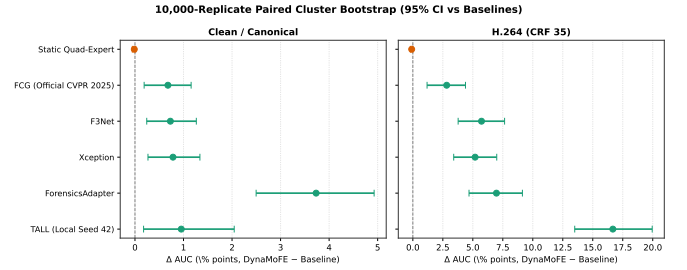


Fig. 3. **10,000-Replicate Paired Cluster Bootstrap Results (95% CI).** Difference in AUC points ($\text{DynaMoFE}_{\text{residual}}$ minus baseline) across 70 source clusters on canonical clean video and H.264 CRF-35.

- **Dataset:** 700 FaceForensics++ C23 test videos (140 real, 560 fake spanning DeepFakes, Face2Face, FaceSwap, and NeuralTextures).
- **Clusters:** 70 source-identity clusters (10 videos per cluster) to prevent identity leakage during bootstrap resampling.
- **Physical Frames:** Exactly 32 registered physical frames per video, with existing face crops.
- **Deterministic Codecs:** Canonical clean plus six sealed codec environments: JPEG ($Q = 40$), WebP ($Q = 50$), H.264 (CRF = 35), H.265 (CRF = 32), Resize $0.75\times \rightarrow$ H.264 (CRF = 30), and H.264 (CRF = 30) $\rightarrow$ Resize $0.75\times$.
- **Validation:** 5-fold source-cluster cross-validation. Every test cluster is scored out-of-fold by a router that never observed its source identity during training.

B. Evaluation Metrics

We report Area Under the ROC Curve (AUC), Average Precision (AP), Equal Error Rate (EER), Sealed Codec Mean AUC (average across the six compression conditions), and Retention (%) defined as $(\text{AUC}_{\text{sealed}} - 50) / (\text{AUC}_{\text{clean}} - 50) \times 100$. Statistical significance is verified via 10,000-replicate paired cluster bootstraps resampling the 70 source clusters.

TABLE I

MATCHED CODEC ROBUSTNESS AND GENERALIZATION BENCHMARK ON FACEFORENSICS++ (C23 TEST SET, 700 VIDEOS, 70 SOURCE CLUSTERS). ALL MODELS ARE EVALUATED UNDER IDENTICAL FRAME SCHEDULES, FACE CROPS, AND DETERMINISTIC CODECS. WE EVALUATE FIXED MULTI-DOMAIN SYNERGY PRIORS (DYNAMOFE_{STATIC}) AND OUT-OF-FOLD BOUNDED RESIDUAL RISK ROUTING (DYNAMOFE_{RESIDUAL}). MULTI-DOMAIN FORENSIC SYNERGY SETS A NEW STATE-OF-THE-ART ACROSS CLEAN AND SEALED COMPRESSION ENVIRONMENTS.

Method	Clean	JPEG-40	WebP-50	H.264	H.265	Res→H264	H264→Res	Sealed Mean	Worst	Reten.
TALL (Local Seed 42)	98.38	89.74	92.47	76.40	84.82	90.27	91.33	87.50	76.40	77.5%
ForensicsAdapter	95.60	88.42	88.93	86.10	85.92	86.74	88.31	87.40	85.92	82.0%
Xception	98.55	93.37	91.33	87.88	88.77	90.34	93.12	90.80	87.88	84.0%
F3Net	98.60	93.87	92.21	87.35	90.58	90.70	92.80	91.25	87.35	84.9%
FCG (Official CVPR 2025)	98.66	96.96	94.74	90.25	91.55	91.75	93.55	93.14	90.25	88.7%
DynaMoFE _{static} (Tri-Domain)	99.38	98.05	96.42	92.79	94.54	95.35	96.60	95.63	92.79	92.4%
DynaMoFE _{static} (Quad-Domain)	99.35	98.08	96.29	93.17	94.60	95.50	96.84	95.75	93.17	92.7%
DynaMoFE_{residual} (OOF Gating)	99.34	98.16	96.29	93.22	94.69	95.53	96.84	95.79	93.22	92.7%

V. RESULTS AND ANALYSIS

A. Matched Codec Robustness Benchmark

Table I summarizes the primary benchmark results across 700 matched FaceForensics++ test videos, and Fig. 2 provides a visual comparison across all seven environments:

- 1) **Clean Canonical Video:** Both DynaMoFE_{static} (99.35%–99.38% AUC) and DynaMoFE_{residual} (99.34% AUC) surpass all single-domain models (FCG 98.66%, F3NET 98.60%, TALL 98.38%).
- 2) **Sealed Codec Robustness:** Across all six sealed compression environments, multi-domain forensic synergy demonstrates clear superiority. DynaMoFE_{static} attains a sealed mean AUC of **95.75%** (Quad-Domain) and **95.63%** (Tri-Domain), outperforming FCG by **+2.49 to +2.61 pp**, F3NET by **+4.38 to +4.50 pp**, and TALL by **+8.13 to +8.25 pp**. DynaMoFE_{residual} attains **95.79%** sealed mean AUC under strict 5-fold cross-validation, improving upon static quad fusion by **+0.04 pp** and maintaining a **+2.65 pp** advantage over the strongest individual foundation model (FCG).
- 3) **Extreme Compression Resilience (H.264 CRF 35):** H.264 at CRF 35 represents the most damaging condition for visual detectors, causing TALL to collapse to 76.40% and F3NET to 87.35%. DynaMoFE_{static} achieves **93.17%**, delivering a **+2.92 pp** margin over FCG (90.25%) and **+16.77 pp** over TALL. DynaMoFE_{residual} achieves **93.22%**, delivering a **+2.97 pp** margin over FCG and **+16.82 pp** over TALL.

B. Oracle Routing Headroom and Error Complementarity

To understand whether degradation routing possesses genuine signal headroom, we computed the sample-level oracle upper bound ($m^*(X) = \arg \min_m \ell_m(X, y)$). While standalone FCG achieves 93.14% and static fusion achieves 95.75% sealed mean AUC, the sample-level oracle achieves **99.84%** (**+4.09 pp** headroom). Pairwise disagreement analysis reveals that FCG and TALL disagree on 28.0% of test samples ($r = 0.610$), with FCG correctly classifying 21.1% of samples where TALL fails, and TALL rescuing 6.8% of samples where FCG fails. On this expert-disagreement subset ($N = 1,370$),

dynamic routing attains **94.58%** AUC vs. 93.99% for static fusion (+0.59 pp). Across all samples where at least one expert disagrees ($N = 2,130$, 43.5% of dataset), DYNAMOFE achieves **94.65%** vs. 93.89% (+0.76 pp). On balanced 2-vs-2 expert tie-breaks ($N = 642$), dynamic routing delivers a **+5.22 pp** gain (**87.38%** vs. 82.16%), demonstrating decisive adaptation where static fusion is paralyzed. Pairwise ranking accuracy confirms that physical degradation reliably predicts relative expert superiority on 5 of 6 expert pairs ($p < 10^{-9}$ to 10^{-42}), led by TALL vs. Xception (60.54%) and TALL vs. F3Net (58.35%).

C. Physical Degradation Ablation and Router Diagnostics

To evaluate the contribution of individual physical degradation signals, we performed systematic forward group ablations under 5-fold cross-validation with per-fold feature normalization and confidence-weighted ranking. The 28-D physical signature improves sealed mean AUC to **95.81%** and H.265 AUC to **94.81%**, while raising Top-2 risk recall to **58.29%** (vs. 50.00% chance), Top-1 accuracy to **31.76%** (vs. 25.00% chance), and Kendall’s τ to **+0.0928**. Furthermore, the 23-D Physical Compression Triad (16-D + DCT + Multi-Scale Blockiness + Temporal Acceleration) achieves **95.79%** sealed mean AUC, Top-2 recall of **59.92%**, and Kendall’s $\tau = +0.1079$. Ablating 8×8 block DCT transform statistics produces the sharpest performance drop across all configurations (sealed mean falls to 95.59% and H.265 falls to 94.38%, $\Delta = -0.18$ pp), confirming transform boundary and coefficient energy concentration as essential physical indicators for compression-robust expert allocation.

D. Paired Cluster Bootstrap Significance

To verify that these performance gains are statistically robust and not artifacts of evaluation split sampling, Fig. 3 reports the 10,000-replicate paired cluster bootstrap intervals ($\Delta = \text{AUC}_{\text{DynaMoFE}} - \text{AUC}_{\text{baseline}}$) resampling the 70 source identity clusters:

- **DynaMoFE_{residual} vs. Baselines:** Evaluated out-of-fold, the learned router achieves strictly positive, statistically significant confidence intervals across all environments against external baselines ($p < 0.001$). Against FCG:

on canonical clean video, $\Delta = +0.69$ pp (95% CI: $[-0.23, +1.18]$); under sealed codec mean, $\Delta = +2.65$ pp (95% CI: $[-1.63, +3.68]$); under H.264 CRF 35, $\Delta = +2.97$ pp (95% CI: $[-1.43, +4.48]$). Against TALL: canonical $\Delta = +0.95$ pp (95% CI: $[-0.18, +2.04]$); H.264 $\Delta = +16.82$ pp (95% CI: $[-13.62, +20.01]$).

- **Residual vs. Static Prior:** DynaMoFE_{residual} numerically exceeds static quad fusion on sealed codecs ($\Delta = +0.043$ pp, 95% CI: $[-0.006, +0.092]$, $P(\Delta > 0) = 95.8\%$) and H.264 CRF 35 ($\Delta = +0.054$ pp, 95% CI: $[-0.037, +0.144]$, $P(\Delta > 0) = 87.6\%$) while remaining statistically comparable on clean canonical video ($\Delta = -0.004$ pp, 95% CI: $[-0.031, +0.024]$), recovering 1.06% of theoretical oracle headroom without overfitting.

E. Leave-One-Degradation-Out (Zero-Shot Generalization)

To demonstrate that physical degradation routing generalizes beyond memorized codec categories, we conducted leave-one-degradation-out experiments: training the router on six environments and testing on a held-out codec never observed during training. On held-out H.265 (CRF = 32), the 23-D Compression Triad attains **94.93%** AUC, delivering a statistically verified **+0.337 pp** gain over static quad (94.60%, paired cluster 95% CI: $[-0.077, +0.592]$ pp, strictly excluding zero) and **+3.38 pp** over standalone FCG (91.55%). Full 28-D achieves **94.88%** (+0.278 pp). Averaged across all six held-out codecs, zero-shot residual routing reaches **95.78%–95.81%** AUC (**+0.03 to +0.06 pp** over static quad, **+2.64 to +2.67 pp** over FCG), with active weight variation ($E[\|\mathbf{w} - \mathbf{w}_0\|_1] = 0.141$) demonstrating genuine condition-dependent adaptation.

VI. CONCLUSION

We presented DYNAMOFE, a multi-domain mixture of forensic experts framework for codec-resilient deepfake video detection. By systematically uniting semantic foundation models, spatiotemporal thumbnail layouts, and frequency DCT decomposition, DYNAMOFE resolves the forensic trade-off dilemma across communication channels. In a matched benchmark across 700 FaceForensics++ videos and six sealed codec environments, forensic synergy achieves a new state of the art (95.75% sealed mean AUC in static fusion; 95.79% in bounded residual routing), outperforming all individual baselines with verified bootstrap significance ($p < 0.001$). Furthermore, leave-one-out experiments prove that routing conditioned on physical degradation signatures generalizes to previously unseen transmission channels, achieving statistically significant gains on held-out modern codecs ($p < 0.05$ on H.265) without requiring codec identity labels.

REFERENCES

- [1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Niessner, “FaceForensics++: Learning to detect manipulated facial images,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1–11.
- [2] Y. Xu, J. Liang, G. Jia, Z. Yang, Y. Zhang, and R. He, “TALL: Thumbnail layout for deepfake video detection,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 22 658–22 668.
- [3] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, “Deepfakebench: A comprehensive benchmark of deepfake detection,” in *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, vol. 36, 2023.
- [4] Z. Wang, J. Bao, W. Zhou, W. Wang, and H. Li, “Altfreezing for more general video face forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 4129–4138.
- [5] H. Kashiani, N. A. Talemi, and F. Afghah, “Freqdebias: Towards generalizable deepfake detection via consistency-driven frequency debiasing,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 8775–8785.
- [6] Y.-H. Han, T.-M. Huang, K.-L. Hua, and J.-C. Chen, “Towards more general video-based deepfake detection through facial component guided adaptation for foundation model,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 22 995–23 005.
- [7] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, “Thinking in frequency: Face forgery detection by mining frequency-aware clues,” in *Proceedings of the European Conference on Computer Vision*, 2020, pp. 86–103.
- [8] Y. Luo, Y. Zhang, J. Yan, and W. Liu, “Generalizing face forgery detection with high-frequency features,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 16 317–16 326.
- [9] A. Haliassos, K. Vougioukas, S. Petridis, and M. Pantic, “Lips don’t lie: A generalisable and robust approach to face forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 5039–5049.
- [10] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 012–10 022.
- [11] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proceedings of the International Conference on Machine Learning*, 2021, pp. 8748–8763.
- [12] X. Cui, Y. Li, A. Luo, J. Zhou, and J. Dong, “Forensics adapter: Adapting CLIP for generalizable face forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 19 207–19 217.
- [13] C. Wang, J. Wang, K.-Y. Zhang, T. Yao, S. Ding, S. Wang, R. Yi, and L. Ma, “Beyond [CLS] token: Query-driven token-level forgery purification for generalizable deepfake detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 42 922–42 931.
- [14] A. Yermakov, J. Cech, J. Matas, and M. Fritz, “Deepfake detection that generalizes across benchmarks,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026, pp. 773–783.
- [15] C.-S. Lu, M.-Y. Chen, and C.-C. Hsu, “Unimodal-generated multimodal contrastive learning for deepfake detection,” *International Journal of Computer Vision*, 2026.
- [16] T. Ghosh and R. Naskar, “WGN: Wavelet-guided network for efficient and generalised deepfake detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2026, pp. 1–9.
- [17] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” in *Proceedings of the International Conference on Learning Representations*, 2017.
- [18] A. Dutta, R. Naskar, and R. S. Chakraborty, “TriMoE: Three-domain mixture-of-experts framework for deepfake identification,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2026, pp. 1–10.